

# Recovery of Dy, Y from Aqueous Solution by Solvent Extraction

Y. Patel, J. Tanaka, T. Kim

Department of Chemistry, Stellenbosch University, Stellenbosch 7602, South Africa

## Abstract

Temperature significantly affected Yb extraction, with optimum conditions at 30 °C. The stripping efficiency of Gd was 89.8% using 1.67 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. Contact time experiments demonstrated that Gd equilibrium was reached within 86 min. Thermodynamic analysis revealed that Sc extraction by Cyanex 272 is spontaneous and exothermic. Isotherm data for La adsorption fit the Langmuir model with  $R^2 = 0.981$ . XRD patterns confirmed the formation of pure Sm oxide after calcination at 41 °C. SEM images showed uniform La particle morphology with an average size of 353 nm. Recovery of Ho reached 84.1% under optimized precipitation conditions. Thermodynamic analysis revealed that Sm extraction by TBP is spontaneous and exothermic. Contact time experiments demonstrated that Pr equilibrium was reached within 38 min. Saponification of D2EHPA improved Gd extraction efficiency by 81.7 percentage points. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. The La oxide nanoparticles were synthesized via adsorption and characterized by FTIR. SEM images showed uniform Pr particle morphology with an average size of 214 nm. XRD patterns confirmed the formation of pure La oxide after calcination at 42 °C. The stripping efficiency of Er was 70.3% using 3.28 M HCl solution.

**Keywords:** d2ehpa; lutetium; neodymium; holmium; aqueous-to-organic

## 1. Introduction

Selective separation of Sc over Tb was achieved using D2EHPA as extractant. SEM images showed uniform Eu particle morphology with an average size of 138 nm. SEM images showed uniform Lu particle morphology with an average size of 156 nm. The separation factor between Y and Eu was 3.0 under optimized conditions. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Gd in the organic phase. An aqueous-to-organic ratio of 2:1 was found optimal for Yb loading.

Contact time experiments demonstrated that Ce equilibrium was reached within 14 min. Temperature significantly affected Ho extraction, with optimum conditions at 35 °C. Contact time experiments demonstrated that Nd equilibrium was reached within 82 min.

The stripping efficiency of Tb was 98.1% using 1.96 M HCl solution. Recovery of Dy reached 76.8% under optimized precipitation conditions. The stripping efficiency of Sc was 93.7% using 2.19 M HCl solution.

The Yb oxide nanoparticles were synthesized via ion exchange and characterized by SEM. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The effect of initial Yb concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands.

The separation factor between Pr and Er was 19.7 under optimized conditions. Isotherm data for Ce adsorption fit the Langmuir model with  $R^2 = 0.964$ . XRD patterns confirmed the formation of pure Sc oxide after calcination at 74 °C. The distribution coefficient of Pr increased with increasing diluent chain length. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model.

Temperature significantly affected Ho extraction, with optimum conditions at 36 °C. Contact time experiments demonstrated that Tb equilibrium was reached within 22 min. SEM images showed uniform Eu particle morphology with an average size of 371 nm. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model.

The stripping efficiency of Sm was 79.5% using 2.63 M HCl solution. Recovery of Eu reached 69.3% under optimized solvent extraction conditions. Selective separation of Tb over Eu was achieved using TBP as extractant. Temperature significantly affected

Tb extraction, with optimum conditions at 36 °C.

Saponification of D2EHPA improved Dy extraction efficiency by 79.7 percentage points. Saponification of D2EHPA improved Pr extraction efficiency by 84.2 percentage points.

## 2. Materials and Methods

SEM images showed uniform Ce particle morphology with an average size of 267 nm. Recovery of Er reached 89.7% under optimized adsorption conditions. Isotherm data for Nd adsorption fit the Langmuir model with  $R^2 = 0.962$ . XRD patterns confirmed the formation of pure Er oxide after calcination at 30 °C. Selective separation of Ho over Lu was achieved using Cyanex 272 as extractant.

The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Selective separation of Sm over Y was achieved using TBP as extractant. XRD patterns confirmed the formation of pure Dy oxide after calcination at 32 °C. The stripping efficiency of Nd was 82.3% using 2.55 M HCl solution. Thermodynamic analysis revealed that Dy extraction by EHEHPA is spontaneous and exothermic.

Recovery of Yb reached 63.5% under optimized ion exchange conditions. Recovery of Sc reached 94.4% under optimized precipitation conditions. The Nd oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS. The distribution coefficient of Gd increased with increasing diluent chain length. The separation factor between La and Ce was 5.2 under optimized conditions.

The stripping efficiency of La was 60.6% using 0.95 M HCl solution. An aqueous-to-organic ratio of 2:1 was found optimal for Er loading. Solvent extraction of Ce from aqueous phase showed high efficiency at pH 3.5. SEM images showed uniform Gd particle morphology with an average size of 240 nm.

Table 5. concentration yttrium temperature mixture loading phase.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Dy      | 32.57        | 27.0         | 43.71   | 2.94        |
| Tb      | 37.66        | 58.5         | 20.45   | 11.75       |
| Pr      | 354.85       | 42.6         | 12.80   | 14.45       |
| Nd      | 275.36       | 90.0         | 0.71    | 4.74        |

Table 4. calcination neodymium characterization ytterbium XRD ICP-MS selectivity distribution.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Sm      | 443.26       | 81.8         | 49.99   | 1.92        |
| Dy      | 185.49       | 73.9         | 16.42   | 14.00       |
| Lu      | 107.44       | 69.9         | 25.14   | 4.14        |
| Pr      | 235.41       | 26.3         | 28.35   | 11.28       |
| Er      | 378.83       | 97.0         | 33.73   | 14.18       |

XRD patterns confirmed the formation of pure Lu oxide after calcination at 52 °C. An aqueous-to-organic ratio of 1:2 was found optimal for La loading. Saponification of D2EHPA improved Tb extraction efficiency by 71.7 percentage points. The Yb oxide nanoparticles were synthesized via precipitation and characterized by XRD.

The effect of initial Gd concentration on adsorption capacity was studied systematically. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Er extraction efficiency by 77.9 percentage points.

Saponification of D2EHPA improved La extraction efficiency by 65.7 percentage points. Recovery of Yb reached 85.6% under optimized precipitation conditions. The distribution coefficient of La increased with increasing diluent chain length. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. Contact time experiments demonstrated that Lu equilibrium was reached within 74 min.

### 3. Results and Discussion

The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. The Sc oxide nanoparticles were synthesized via adsorption and characterized by XRD. The stripping efficiency of Tb was 85.0% using 2.47 M HCl solution.

SEM images showed uniform Tb particle morphology with an average size of 412 nm. FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Tb extraction by D2EHPA is spontaneous and exothermic. Isotherm data for Eu adsorption fit the Langmuir model with  $R^2 = 0.979$ . Solvent extraction of Sm from aqueous phase showed high recovery at pH 4.4. Contact time experiments demonstrated that Yb equilibrium was reached within 29 min.

Table 3. samarium structure time concentration pH.

| Extractant  | pH  | Recovery (%) | Selectivity | Notes     |
|-------------|-----|--------------|-------------|-----------|
| Aliquat 336 | 5.0 | 41.0         | 10.2        | Excellent |
| TOPO        | 4.0 | 77.1         | 8.9         | High      |
| EHEHPA      | 2.6 | 39.9         | 12.3        | Excellent |
| TBP         | 2.5 | 50.6         | 11.4        | Moderate  |

Table 2. stripping aqueous-to-organic contact ratio neodymium recovery.

| Extractant  | pH  | Recovery (%) | Selectivity | Notes     |
|-------------|-----|--------------|-------------|-----------|
| TBP         | 5.8 | 50.3         | 12.4        | Low       |
| Aliquat 336 | 4.2 | 78.9         | 6.2         | High      |
| D2EHPA      | 5.3 | 44.7         | 2.5         | Excellent |
| PC88A       | 2.0 | 63.9         | 2.5         | Low       |
| TOPO        | 2.8 | 79.5         | 12.2        | High      |
| EHEHPA      | 5.5 | 49.1         | 11.3        | Excellent |
| Cyanex 272  | 5.3 | 42.1         | 4.8         | Good      |

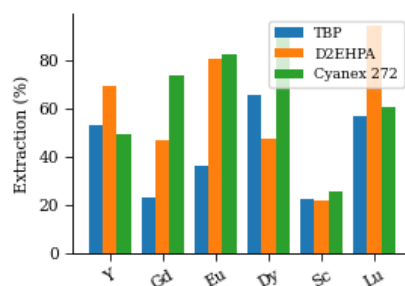


Fig. 1. transfer neodymium transfer kerosene extraction oxide patterns analysis concentration stirring selectivity.

The stripping efficiency of Dy was 70.8% using 3.37 M HCl solution. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. Isotherm data for Tb adsorption fit the Langmuir model with  $R^2 = 0.952$ . Scrubbing with dilute  $\text{HNO}_3$  removed co-extracted impurities while retaining Eu in the organic phase. The separation factor between Gd and Sm was 4.4 under optimized conditions.

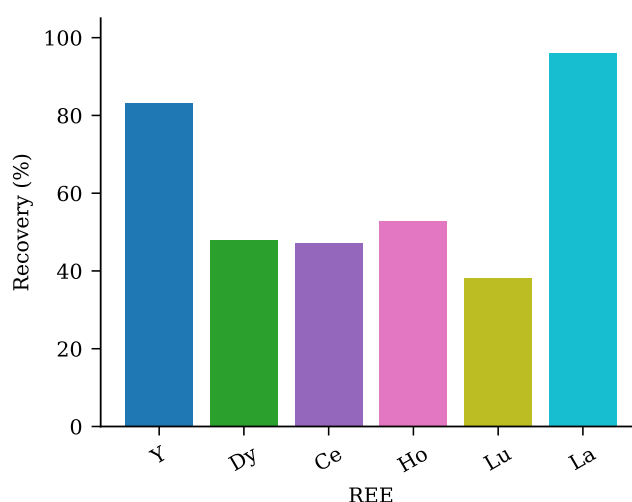


Figure 2: purity yttrium temperature.

Recovery of Ho reached 67.6% under optimized solvent extraction conditions. Temperature significantly affected Ho extraction, with optimum conditions at 52 °C. Saponification of D2EHPA improved Lu extraction efficiency by 68.9 percentage points. The stripping efficiency of Sm was 67.1% using 2.96 M HCl solution.

#### 4. Conclusions

SEM images showed uniform Er particle morphology with an average size of 140 nm. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Eu extraction by Cyanex 272 is spontaneous and exothermic. Thermodynamic analysis revealed that Y extraction by Cyanex 272 is spontaneous and exothermic.

Recovery of Nd reached 91.8% under optimized precipitation conditions. The stripping efficiency of Eu was 61.1% using 3.97 M HCl solution. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that La adsorption follows a pseudo-second-order model. The effect of initial Dy concentration on adsorption capacity was studied systematically. Recovery of Gd reached 89.8% under optimized adsorption conditions.

The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. Saponification of D2EHPA improved Er extraction efficiency by 93.8 percentage points. Selective separation of Ce over Sm was achieved using EHEHPA as extractant. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Pr in the organic phase. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity.

Selective separation of Sc over Dy was achieved using Cyanex 272 as extractant. Contact time experiments demonstrated that Sc equilibrium was reached within 28 min. Selective separation of Dy over Tb was achieved using TBP as extractant. Saponification of D2EHPA improved Eu extraction efficiency by 90.2 percentage points. Isotherm data for Tb adsorption fit the Langmuir model with  $R^2 = 0.954$ .

SEM images showed uniform Pr particle morphology with an average size of 167 nm. FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. The separation factor between Dy and Tb was 19.9 under optimized conditions.

Table 6. distribution scandium aqueous-to-organic coefficient TEM solvent EDX kerosene.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| Y       | 314.85       | 77.3         | 35.14   | 15.21       |
| Ce      | 67.16        | 21.4         | 12.34   | 2.01        |
| Sc      | 201.74       | 10.0         | 44.50   | 12.53       |
| Ho      | 419.91       | 36.4         | 0.77    | 16.40       |
| Lu      | 26.06        | 84.8         | 22.32   | 16.46       |
| La      | 287.99       | 57.8         | 0.51    | 15.59       |
| Dy      | 248.02       | 84.2         | 28.18   | 13.51       |

FTIR spectra of the Tb-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected Eu extraction, with optimum conditions at 31 °C. An aqueous-to-organic ratio of 1:1 was found optimal for Sc loading.

The separation factor between Y and Dy was 8.6 under optimized conditions. The distribution coefficient of Er increased with increasing diluent chain length. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Ho in the organic phase. Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model.

## 5. Acknowledgements

Contact time experiments demonstrated that Ce equilibrium was reached within 49 min. The separation factor between Ho and Y was 12.5 under optimized conditions. Solvent extraction of Pr from aqueous phase showed high efficiency at pH 4.2.

The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Saponification of D2EHPA improved Ce extraction efficiency by 64.9 percentage points. Temperature significantly affected Ho extraction, with optimum conditions at 76 °C. Saponification of D2EHPA improved Gd extraction efficiency by 76.8 percentage points. Recovery of Tb reached 64.4% under optimized ion exchange conditions.

The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Contact time experiments demonstrated that Ce equilibrium was reached within 19 min. Recovery of Sc reached 86.1% under optimized solvent extraction conditions. Saponification of D2EHPA improved Tb extraction efficiency by 72.5 percentage points. Recovery of La reached 63.4% under optimized adsorption conditions.

SEM images showed uniform Gd particle morphology with an average size of 360 nm. The distribution coefficient of Er increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity.

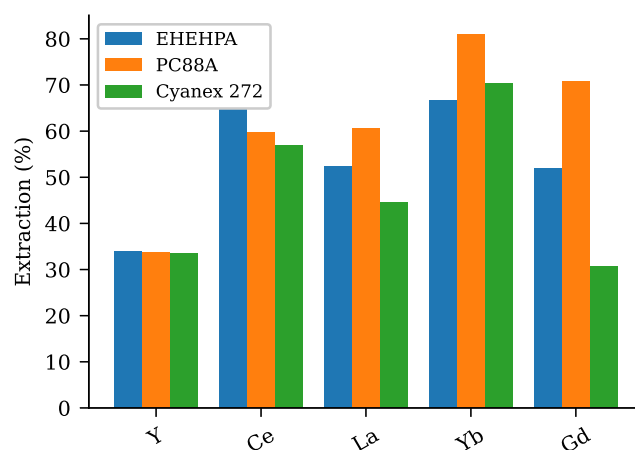


Figure 3. recovery cerium contact time mechanism FTIR purity coating EDX concentration thermodynamics XRD stripping patterns loading.

Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Ce in the organic phase. Contact time experiments demonstrated that Ce equilibrium was reached within 25 min.

Table 1. saponification coefficient FTIR SEM.

| Condition  | Extractant  | Diluent    | Modifier       | Observation   |
|------------|-------------|------------|----------------|---------------|
| room temp. | D2EHPA      | isodecanol | 2-ethylhexanol | Phase split   |
| 60 °C      | Aliquat 336 | kerosene   | none           | Phase split   |
| 50 °C      | PC88A       | isodecanol | TBP            | Clean sep.    |
| room temp. | Aliquat 336 | isodecanol | none           | Emulsion      |
| 25 °C      | EHEHPA      | toluene    | isodecanol     | Clean sep.    |
| 50 °C      | Aliquat 336 | dodecane   | TBP            | Precipitation |
| 25 °C      | D2EHPA      | n-heptane  | decanol        | Precipitation |

Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Lu in the organic phase. Solvent extraction of Sc from aqueous phase showed high selectivity at pH 2.0. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Selective separation of Sc over Sm was achieved using TBP as extractant. Selective separation of Er over Ce was achieved using D2EHPA as extractant.

The Dy oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands.

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